

# Validated AI Threat Taxonomy

Threat-focused taxonomy for Traditional AI, LLM, Agentic AI, and AI-enabled threats - June 2026 revision.

**Revision focus:** this version fixes three issues from the previous audit: it removes non-threat items from the primary tag layer, strengthens the agentic runtime/infrastructure layer, and replaces broad AI-enabled mappings with paired ATT&CK; technique plus AI capability modifiers.

## Design Rules

| Rule                                  | Meaning                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Threat tags only                      | Primary tags describe adversarial behaviour, attack paths, abuse mechanisms, or materially security-relevant capabilities.                |
| No generic risk tags                  | Items such as misinformation, overreliance, and excessive agency are treated as impacts or enabling conditions, not primary threat tags.  |
| Agentic infrastructure included       | Agentic AI is modelled across memory, context, tools, MCP, runtime, credentials, workflows, orchestration, and multi-agent communication. |
| AI-enabled threats use paired mapping | Each AI-enabled tag maps to a cyber technique or operation plus an AI capability modifier, instead of using T1588.007 as a catch-all.     |
| Validation requirement                | A tag should only be assigned when source evidence directly supports the threat behaviour and includes a traceable source URL.            |

## Primary Taxonomy

### Traditional AI Threats

Primary adversarial ML and AI-system compromise threats. Retained as MITRE ATLAS-centred tags.

| Tag                        | Threat meaning                                                                      | Reference                         | Source              |
|----------------------------|-------------------------------------------------------------------------------------|-----------------------------------|---------------------|
| data_poisoning             | Poison training or operational data to alter model behaviour.                       | MITRE ATLAS AML.T0020             | MITRE ATLAS         |
| model_poisoning            | Alter or implant malicious behaviour in model artefacts or training pipeline.       | MITRE ATLAS AML.T0018 / AML.T0020 | MITRE ATLAS         |
| adversarial_evasion        | Craft inputs to evade or mislead an AI model at inference.                          | MITRE ATLAS AML.T0015             | MITRE ATLAS         |
| adversarial_data           | Craft adversarial data to exploit model weaknesses.                                 | MITRE ATLAS AML.T0043             | MITRE ATLAS         |
| model_extraction           | Extract model behaviour, weights, or decision boundaries via access or queries.     | MITRE ATLAS AML.T0024 / AML.T0048 | MITRE ATLAS         |
| model_inversion            | Infer sensitive training data or attributes from model outputs.                     | MITRE ATLAS AML.T0024 / AML.T0053 | MITRE ATLAS         |
| membership_inference       | Infer whether a record was present in training data.                                | MITRE ATLAS AML.T0024             | MITRE ATLAS         |
| inference_api_abuse        | Abuse model inference API access for extraction, evasion, leakage, or cost attack.  | MITRE ATLAS AML.T0040             | MITRE ATLAS         |
| model_denial_of_service    | Degrade availability or cost profile of AI model service.                           | MITRE ATLAS AML.T0029 / AML.T0034 | MITRE ATLAS SAFE-AI |
| model_integrity_erosion    | Degrade model integrity or reliability through adversarial manipulation.            | MITRE ATLAS AML.T0031             | MITRE ATLAS         |
| ai_supply_chain_compromise | Compromise datasets, models, repositories, dependencies, or ML pipeline components. | MITRE ATLAS AML.T0010             | MITRE ATLAS         |

### LLM Threats - Primary Threat Tags

Non-threat items from the previous version have been removed from this primary tag layer and placed in the secondary dimensions table.

| Tag                       | Threat meaning                                                                   | Reference                          | Source                |
|---------------------------|----------------------------------------------------------------------------------|------------------------------------|-----------------------|
| prompt_injection          | Manipulate an LLM with malicious instructions.                                   | OWASP LLM01; MITRE ATLAS AML.T0051 | OWASP LLM Top 10 2025 |
| direct_prompt_injection   | User-supplied prompt directly manipulates model behaviour.                       | MITRE ATLAS AML.T0051              | MITRE ATLAS SAFE-AI   |
| indirect_prompt_injection | Malicious instructions are embedded in content retrieved or consumed by the LLM. | MITRE ATLAS AML.T0051; OWASP LLM01 | MITRE ATLAS SAFE-AI   |

| Tag                              | Threat meaning                                                                           | Reference                                 | Source                |
|----------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------|
| multimodal_prompt_injection      | Prompt injection through image, audio, document, or other non-text modality.             | OWASP LLM01                               | OWASP LLM Top 10 2025 |
| rag_prompt_injection             | Prompt injection through retrieval-augmented generation content or knowledge base.       | MITRE ATLAS AML.T0070; OWASP LLM01/LLM08  | MITRE ATLAS           |
| jailbreak                        | Bypass model safeguards or restrictions to elicit disallowed behaviour.                  | MITRE ATLAS AML.T0054                     | MITRE ATLAS           |
| guardrail_bypass                 | Bypass content filters, safety controls, or policy layers around an LLM.                 | MITRE ATLAS AML.T0054                     | MITRE ATLAS           |
| sensitive_information_disclosure | Expose confidential data, PII, secrets, or protected system details.                     | OWASP LLM02                               | OWASP LLM Top 10 2025 |
| system_prompt_leakage            | Extract hidden instructions, system prompt, or private policy text.                      | OWASP LLM07; MITRE ATLAS AML.T0056        | OWASP LLM Top 10 2025 |
| context_leakage                  | Leak sensitive context-window data or conversation state.                                | OWASP LLM02                               | OWASP LLM Top 10 2025 |
| llm_supply_chain_compromise      | Compromise LLM models, datasets, fine-tunes, services, dependencies, or integrations.    | OWASP LLM03/LLM04                         | OWASP LLM Top 10 2025 |
| improper_output_handling         | Unsafe use of LLM output enables injection, SSRF, XSS, code execution, or logic abuse.   | OWASP LLM05                               | OWASP LLM Top 10 2025 |
| vector_embedding_weaknesses      | Exploit embeddings or vector search to leak, poison, or retrieve unintended information. | OWASP LLM08                               | OWASP LLM Top 10 2025 |
| vector_database_exposure         | Expose or abuse vector stores, indexes, metadata, or retrieval permissions.              | OWASP LLM08                               | OWASP LLM Top 10 2025 |
| unbounded_consumption            | Cause resource exhaustion or excessive inference cost.                                   | OWASP LLM10                               | OWASP LLM Top 10 2025 |
| model_theft                      | Steal model capability, weights, or proprietary behaviour.                               | MITRE ATLAS AML.T0024 / OWASP model theft | MITRE ATLAS           |

## Agentic AI Threats - Runtime, Tooling, MCP, Workflow, and Multi-Agent Threats

This fixes the previous infrastructure gap. Tags focus on observable threats against agent memory, context, tools, credentials, runtime, orchestration, and communication.

| Tag                             | Threat meaning                                                                                         | Reference                                           | Subdomain      |
|---------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------|----------------|
| agent_prompt_injection          | Manipulate agent reasoning or action selection through malicious instructions.                         | OWASP LLM01; MITRE ATLAS AML.T0051                  | prompt/control |
| indirect_agent_prompt_injection | Poison webpages, documents, emails, tool outputs, or retrieved content consumed by an agent.           | OWASP LLM01; MITRE ATLAS AML.T0051                  | prompt/control |
| agent_memory_poisoning          | Poison persistent or session memory to alter future agent decisions.                                   | OWASP Agentic AI T1                                 | memory/state   |
| agent_context_poisoning         | Manipulate contextual state, retrieved context, or prompt-state to influence agent behaviour.          | OWASP MCP Top 10; OWASP Agentic AI                  | memory/state   |
| memory_exfiltration             | Use agent memory or context mechanisms to leak sensitive information.                                  | OWASP Agentic AI T1; OWASP LLM02                    | memory/state   |
| tool_poisoning                  | Compromise tool descriptions, schemas, outputs, or registry entries to manipulate model actions.       | OWASP MCP03:2025                                    | tools/MCP      |
| tool_misuse                     | Cause an agent to invoke legitimate tools in unsafe or unintended ways.                                | OWASP Agentic AI T2                                 | tools/MCP      |
| tool_hijacking                  | Redirect or abuse tool calls, plugins, APIs, or MCP endpoints.                                         | OWASP Agentic AI T2; OWASP MCP Top 10               | tools/MCP      |
| mcp_server_compromise           | Abuse or compromise MCP servers that expose tools, context, credentials, or actions.                   | OWASP MCP Top 10; OWASP MCP Security Cheat Sheet    | tools/MCP      |
| tool_supply_chain_compromise    | Compromise agent tools, MCP servers, plugins, dependencies, or skill packages.                         | OWASP MCP04; OWASP LLM03                            | tools/MCP      |
| agent_privilege_abuse           | Exploit excessive agent permissions, delegated authority, or downstream access.                        | OWASP Agentic AI T3; OWASP LLM06                    | permissions    |
| agent_credential_abuse          | Steal, misuse, or overuse tokens, OAuth grants, API keys, or service credentials accessible to agents. | OWASP MCP Security Cheat Sheet; OWASP Agentic AI T3 | permissions    |
| unsafe_code_execution           | Manipulate agent-generated or agent-executed code to perform unintended operations.                    | OWASP Agentic AI T11; OWASP LLM05                   | execution      |
| sandbox_escape                  | Break out of intended agent or code execution containment.                                             | OWASP Agentic AI T11                                | execution      |
| workflow_poisoning              | Manipulate autonomous workflow steps, queues, plans, or task chains.                                   | OWASP Agentic AI; OWASP Agentic Skills Top 10       | workflow       |
| orchestration_compromise        | Compromise planner, router, coordinator, or multi-step orchestration layer.                            | OWASP Agentic AI; OWASP Agentic Skills Top 10       | workflow       |
| agent_communication_poisoning   | Manipulate messages or shared state between agents.                                                    | OWASP Agentic AI T12                                | multi-agent    |
| multi_agent_coordination_abuse  | Exploit multi-agent collaboration to amplify, hide, or cascade harmful actions.                        | OWASP Agentic AI T13                                | multi-agent    |
| agent_identity_spoofing         | Impersonate agent, user, role, or authority boundary.                                                  | OWASP Agentic AI T9                                 | identity       |
| human_agent_manipulation        | Use agent behaviour or outputs to manipulate human operators into unsafe actions.                      | OWASP Agentic AI T10/T15                            | human-agent    |

## AI-Enabled Threats - Paired Operational Mapping

This fixes ATT&CK; overloading by separating the operational technique from the AI capability modifier. T1588.007 is a modifier, not the only reference.

| Tag                                | Threat meaning                                                                           | Operational mapping    | AI modifier               |
|------------------------------------|------------------------------------------------------------------------------------------|------------------------|---------------------------|
| ai_assisted_reconnaissance         | Reconnaissance and OSINT accelerated or improved by AI.                                  | T1595 / Reconnaissance | T1588.007 - AI capability |
| ai_target_profiling                | AI-assisted profiling of individuals, roles, organisations, or identities for targeting. | T1589                  | T1588.007 - AI capability |
| ai_assisted_vulnerability_research | AI helps identify, interpret, rank, or operationalise vulnerability information.         | T1588.006              | T1588.007 - AI capability |
| ai_exploit_development             | AI helps develop, adapt, test, or explain exploits.                                      | T1587.004              | T1588.007 - AI capability |
| ai_malware_development             | AI helps develop, modify, or assemble malware or malicious tooling.                      | T1587.001              | T1588.007 - AI capability |
| ai_payload_obfuscation             | AI assists obfuscation, mutation, evasion, or transformation of payloads/scripts.        | T1027                  | T1588.007 - AI capability |

| Tag                         | Threat meaning                                                                             | Operational mapping                           | AI modifier                                  |
|-----------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------|
| ai_command_generation       | AI generates commands, scripts, or operator instructions for intrusion activity.           | T1059                                         | T1588.007 - AI capability                    |
| ai_assisted_phishing        | AI generates, personalises, translates, or scales phishing and social engineering.         | T1566                                         | T1588.007 - AI capability                    |
| ai_voice_impersonation      | Synthetic voice is used for vishing, fraud, or trusted-person impersonation.               | T1566 / social engineering                    | T1588.007 - AI-generated audio               |
| ai_deepfake_impersonation   | Synthetic image, video, or audio impersonates a trusted identity.                          | T1566 / social engineering                    | T1588.007 - AI-generated media               |
| synthetic_identity_abuse    | AI-generated identities support deception, fraud, account creation, or access attempts.    | T1589 / identity information                  | T1588.007 - generated identity material      |
| ai_enabled_fraud            | AI materially enables fraud, BEC, scam operations, or impersonation-based financial abuse. | T1566 / fraud use case                        | T1588.007 - AI social engineering            |
| ai_generated_disinformation | AI-generated content supports influence, manipulation, or disinformation operations.       | DISARM Red Framework                          | T1588.007 - generated text/image/audio/video |
| ai_attack_automation        | AI automates or coordinates repeated steps across an attack workflow.                      | Relevant ATT&CK; technique required per case  | T1588.007 - AI automation modifier           |
| ai_orchestrated_intrusion   | AI is used to plan, coordinate, or accelerate multiple intrusion phases.                   | Relevant ATT&CK; techniques required per case | T1588.007 - AI orchestration modifier        |

## Secondary Dimensions - Not Primary Threat Tags

These labels may be useful for analytics and synthesis, but they should not be counted as primary threat techniques. This resolves the previous problem where failure modes and impacts were mixed with threats.

| Label                   | Dimension                 | Use rule                                                                                                            | Reference                         |
|-------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| excessive_agency        | enabling_condition        | Use when excessive permissions/autonomy enable a threat; do not count as attack technique by itself.                | OWASP LLM06 / OWASP Agentic AI T3 |
| misinformation          | impact                    | Use as an impact when false outputs or generated content are the result of a threat.                                | OWASP LLM09                       |
| overreliance            | control_failure           | Use when humans or systems over-trust AI output; not an adversarial technique.                                      | OWASP LLM09                       |
| resource_overload       | impact/enabling_condition | Use as effect or stress condition; primary threat tag should be unbounded_consumption or agent_resource_exhaustion. | OWASP LLM10 / OWASP Agentic AI T4 |
| cascading_hallucination | failure_mode              | Use as failure mode in agent chains, not primary threat unless adversarially induced.                               | OWASP Agentic AI T5               |

## Reference Links

The links below were selected as primary framework references for this taxonomy. Use source-specific evidence in the pipeline in addition to these framework mappings.

| Reference                                | URL                                                                                                                                                                           |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MITRE ATLAS                              | <a href="https://atlas.mitre.org/">https://atlas.mitre.org/</a>                                                                                                               |
| MITRE ATLAS SAFE-AI                      | <a href="https://atlas.mitre.org/pdf-files/SAFEAI_Full_Report.pdf">https://atlas.mitre.org/pdf-files/SAFEAI_Full_Report.pdf</a>                                               |
| OWASP LLM Top 10 2025                    | <a href="https://owasp.org/www-project-top-10-for-large-language-model-applications/">https://owasp.org/www-project-top-10-for-large-language-model-applications/</a>         |
| OWASP Agentic AI Threats and Mitigations | <a href="https://genai.owasp.org/resource/agentic-ai-threats-and-mitigations/">https://genai.owasp.org/resource/agentic-ai-threats-and-mitigations/</a>                       |
| OWASP MCP Top 10                         | <a href="https://owasp.org/www-project-mcp-top-10/">https://owasp.org/www-project-mcp-top-10/</a>                                                                             |
| OWASP MCP Tool Poisoning                 | <a href="https://owasp.org/www-project-mcp-top-10/2025/MCP03-2025%E2%80%93Tool-Poisoning">https://owasp.org/www-project-mcp-top-10/2025/MCP03-2025%E2%80%93Tool-Poisoning</a> |
| OWASP MCP Security Cheat Sheet           | <a href="https://cheatsheetseries.owasp.org/cheatsheets/MCP_Security_Cheat_Sheet.html">https://cheatsheetseries.owasp.org/cheatsheets/MCP_Security_Cheat_Sheet.html</a>       |
| MITRE ATT&CK; T1588.007 AI               | <a href="https://attack.mitre.org/techniques/T1588/007/">https://attack.mitre.org/techniques/T1588/007/</a>                                                                   |
| MITRE ATT&CK; T1587.001 Malware          | <a href="https://attack.mitre.org/techniques/T1587/001/">https://attack.mitre.org/techniques/T1587/001/</a>                                                                   |
| MITRE ATT&CK; T1587.004 Exploits         | <a href="https://attack.mitre.org/techniques/T1587/004/">https://attack.mitre.org/techniques/T1587/004/</a>                                                                   |
| MITRE ATT&CK; T1588.006 Vulnerabilities  | <a href="https://attack.mitre.org/techniques/T1588/006/">https://attack.mitre.org/techniques/T1588/006/</a>                                                                   |
| MITRE ATT&CK; T1566 Phishing             | <a href="https://attack.mitre.org/techniques/T1566/">https://attack.mitre.org/techniques/T1566/</a>                                                                           |

| Reference                                              | URL                                                                                                 |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| MITRE ATT&CK; T1027 Obfuscated Files or Information    | <a href="https://attack.mitre.org/techniques/T1027/">https://attack.mitre.org/techniques/T1027/</a> |
| MITRE ATT&CK; T1059 Command and Scripting Interpreter  | <a href="https://attack.mitre.org/techniques/T1059/">https://attack.mitre.org/techniques/T1059/</a> |
| MITRE ATT&CK; T1589 Gather Victim Identity Information | <a href="https://attack.mitre.org/techniques/T1589/">https://attack.mitre.org/techniques/T1589/</a> |
| MITRE ATT&CK; T1595 Active Scanning                    | <a href="https://attack.mitre.org/techniques/T1595/">https://attack.mitre.org/techniques/T1595/</a> |
| DISARM Red Framework                                   | <a href="https://www.disarm.foundation/framework">https://www.disarm.foundation/framework</a>       |

**Validation note:** A framework reference validates the existence and relevance of a threat class. It does not validate any individual factual claim in a report. Individual claims still require source-level evidence, source URLs, and confidence scoring.